

MATLAB Lab Manual - Classroom

matlab.mathworks.com

HOME PLOTS APPS FIGURE FORMAT TOOLS

Save As Copy Figure Print Show Code FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

MATLAB Drive

```
untitled.m x +
/MATLAB Drive/untitled.m
1 clc;
2 clear;
3 close all;
4 antennas = [2 4 8 16 32]; % Number of Antenna Elements
5 gain = 10*log10(antennas); % Beamforming Gain (dB)
6 figure
7 bar(antennas, gain)
8 grid on
9
10 xlabel('Number of Antenna Elements')
11 ylabel('Beamforming Gain (dB)')
12 title('Beamforming Gain for Different Antenna Array Sizes')
```

Figure 1 x

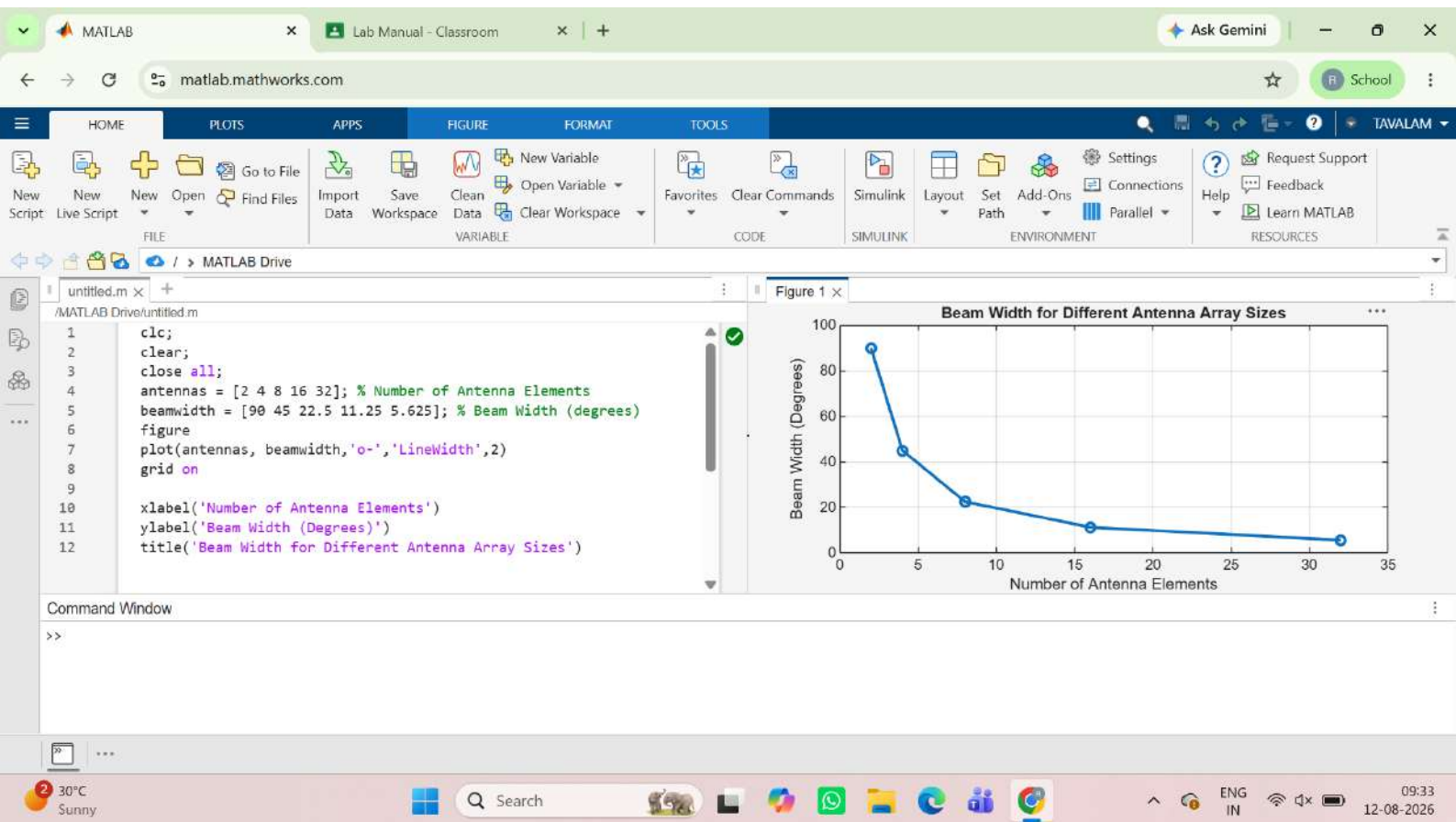
Beamforming Gain for Different Antenna Array Sizes

Number of Antenna Elements	Beamforming Gain (dB)
2	3.01
4	6.02
8	9.03
16	12.04
32	15.05

Command Window

```
>>
```

30°C Sunny Search 09:32 12-08-2026



MATLAB Lab Manual - Classroom

matlab.mathworks.com

HOME PLOTS APPS FIGURE FORMAT TOOLS

New Script New Live Script New Open Go to File Find Files Import Data Save Workspace Clean Data Open Variable Clear Workspace Favorites Clear Commands Simulink Layout Set Path Add-Ons Connections Help Request Support Feedback Learn MATLAB

FILE VARIABLE CODE SIMULINK ENVIRONMENT RESOURCES

/ > MATLAB Drive

```
untitled.m x +
1 clc;
2 clear;
3 close all;
4 theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
5 gain = [1 1 1 1 1]; % Constant normalized gain
6 figure
7 polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
8 title('Main Lobe Direction in Analog Beamforming')
9
```

Figure 1 x

Main Lobe Direction in Analog Beamforming

Command Window

>>

30°C Sunny 09:34 12-08-2026